a Visible writing and conditional state

Example raster cells

Causal visual reader

60 images; frozen 16.28M

Continuous condition

hidden state + visual field

b Joint future-image training

Noisy future strip

four 32 × 32 cells

Conditional U-Net

velocity; FiLM; cell attention

Learning signal

velocity + image losses

Stratified cell noise alternates with synchronous noise

c Raster generation and visible feedback

Gaussian noise

32-level reverse process

fixed guidance 2; first sample

Actual generated raster